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Department of Aeronautical Engineering

2023-2024 EVEN SEM

(2021-25 Batch)

VALUE ADDED COURSE

“Nano Satellites”

08.01.2024 - 12.01.2024


 COORDINATOR


 H.O.D


 PRINCIPAL
 PRINCIPAL
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Department of Aeronautical Engineering Academic Year 2023-2024 EVEN Semester

Department Circular	Date: 26.12.2023
Issuing Authority: HOD	Classification: Periodical
Subject : Value added course -Reg	

We are planning to conduct the Value Added Course **Nano Satellites** in association with Jet Aerospace Research Centre for third year Aeronautical students. Hence all the third year students are informed to register their names to the respective class coordinators for the value added course is going to be conducted on 08.01.2024-12.01.2024 (5 Days) during 9.30 am to 4.30 pm.(Total No of Hours: 40 Hours)

Topics to be covered:

Nano Satellites

Faculty In-Charge:

Mr.Ramana.S/ AP/ Aero

Schedule:

Venue: AB 213, MAMSE.

Course Code & Name: NS - Nano Satellites	
S. No	Topics to be covered
1.	Introduction to Nano Satellites
2.	Satellite Systems and Subsystems
3.	Mission Planning and Design
4.	Development and Testing
5.	Operations and Data Analysis

Circular to:

- All Staff members of Aeronautical,
- To be read in III Year class,
- Copy to the Notice Board.

HOD/AERO

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Department of Aeronautical Engineering

Organizes
Value Added Course on

Nano Satellites

In collaboration with
Jet Aerospace
Research Centre



Date : From 08.01.2024 to 12.01.2024 Time : From 09.30 AM to 04.30 PM

Mr.S.Ramana
Coordinator

Dr.K.Chandrasekaran
Head of the Department

Dr.P.Ranjith Kumar
Principal

NBA - C9 - Co Curricular & Extra Curricular Activities

NAAC - C5 - Co Curricular Activities

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Department of Aeronautical Engineering

Academic Year 2023-2024 Even Semester

List of Participants for Value Added Course

NS – Nano Satellites

Date: 08.01.2024-12.01.2024

S. No	Reg. No	Name	Signature	S. No	Reg. No	Name	Signature
1	812121101002	ABINAYA E		25	812121101031	MANIKANDAN M	
2	812121101004	ABITHRA G		26	812121101032	MADHANKUMAR	
3	812121101005	AKASH A		27	812121101033	MOHD MEERAN.M	
4	812121101006	ALBERT A		28	812121101034	MUTHU KRISHNAN T	
5	812121101007	ARCHANA R		29	812121101035	NAVEEN KUMAR.L	
6	812121101010	ATCHAYA M		30	812121101036	NIRANJANEE.R	
7	812121101011	ATCHAYA.S		31	812121101037	NIVETHA V	
8	812121101012	BALASUBRAMANIAN.M		32	812121101038	OMANATHAN.G.L	
9	812121101013	BOOBALAN J		33	812121101039	PARTHIBAN B	
10	812121101014	DIHANUSIL.D		34	812121101040	POOVARASAN K	
11	812121101015	DHATCHANAMOORTHY A		35	812121101041	PRAVEENA J	
12	812121101016	EDWIN JOSWA		36	812121101043	SAKTHIVEL M	
13	812121101017	ELAVARASU S		37	812121101044	SATHISH V	
14	812121101018	GOPINATH D		38	812121101045	SELVAKUMAR P	
15	812121101019	HAJI MOHAMED S		39	812121101046	SHANMUGAM	
16	812121101020	HARISH B		40	812121101047	SIVAMURUGAN M	
17	812121101022	JEYA PRIYA.G		41	812121101048	SNEKA.K	
18	812121101023	JAYAVEERA PANDI		42	812121101050	SYED SABITHULLA S	
19	812121101024	JITHENDRAA P		43	812121101052	VIGNESH S	
20	812121101025	JOTHIBOSU		44	812121101053	VIJAYALAKSHMI	
21	812121101026	KALAIYARASAN		45	812121101054	VIRUMANDI M	
22	812121101027	KALIDASS		46	812121101055	VISHALLV	
23	812121101029	KAVIYARASAN.A		47	812121101056	VISHNU D	
24	812121101030	LEO STANISLAUS		48	812121101058	YUVANESHWARAN	

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Department of Aeronautical Engineering
Academic Year 2023-2024 (Even SEMESTER)

Date: 08.01.2024

Content of the Program: Nano Satellites

Module 1: Introduction to Nano Satellites

1. Definition and Classification
 - Nano, micro, and pico satellites
 - Historical development and milestones
2. Applications and Benefits
 - Earth observation
 - Communication & Scientific research
 - Education and technology demonstration

Module 2: Satellite Systems and Subsystems

1. Structure and Design
 - Structural components and materials
 - Design considerations and constraints
2. Power Systems & Communication Systems
 - Solar panels and batteries
 - Power management and distribution
 - Antennas and transceivers
 - Communication protocols and frequencies

Module 3: Mission Planning and Design

1. Mission Requirements and Objectives
 - Defining mission goals and success criteria
 - Stakeholder analysis and engagement
2. Orbit Selection
 - Low Earth Orbit (LEO) characteristics
 - Orbit determination and prediction



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Module 4: Development and Testing

1. Design and Development Process

- Systems engineering approach
- Design reviews and iterations

2. Simulation and Modeling

- Software tools for satellite design
- Environmental simulations (thermal, vibration, radiation)
- Testing and Validation

Module 5: Operations and Data Analysis

1. Launch and Deployment

- Launch preparations and procedures
- Deployment mechanisms and strategies

2. Satellite Operations

- Ground stations and network operations
- Command and control procedures
- Anomaly detection and resolution

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Course Plan & Schedule

Course Code: NS

Course Name: Nano Satellites

Date	Session	Hours	Title
08.01.2024	F.N	4 hrs	Introduction to Nano Satellites
	A.N	4 hrs	Applications and Benefits
09.01.2024	F.N	4 hrs	Structure and Design
	A.N	4 hrs	Power Systems & Communication Systems
10.01.2024	F.N	4 hrs	Mission Requirements and Objectives
	A.N	4 hrs	Orbit Selection
11.01.2024	F.N	4 hrs	Design and Development Process
	A.N	4 hrs	Simulation and Modeling
12.01.2024	F.N	4 hrs	Operations and Data Analysis
	A.N	4 hrs	Project and Assessment
Total: 6 Days	10 Sessions	40 Hours	

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Snapshots of Sessions





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VAC – EVALUATION SHEET

Name of the Student : Dhanush. D
Register Number : 812121101014
Sem / Year / Dept : VI Sem / III Year AERO

Mark Allotment			
S. No	Description	Allotted Marks	Obtained Marks
1	Quiz	20	18
2	Group Discussion	20	18
3	Model Making	50	46
4	Assignment	10	9
	TOTAL	100	91

Grade: Excellent


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NS – Nano Satellites

VALUE ADDED COURSE - EVALUATION SHEET

S. No	Reg. No	Name	QUIZ	GD	MODEL MAKING	ASSIGNMENT	Total Marks	Grade
			20	20	50	10	100	
1	812121101002	ABINAYA E	18	18	46	9	91	Excellent
2	812121101004	ABITHRA	15	15	37	7	74	Good
3	812121101005	AKASH A	16	16	40	8	80	Very Good
4	812121101006	ALBERT A	16	16	41	8	81	Very Good
5	812121101007	ARCHANA R	18	18	46	9	91	Excellent
6	812121101010	ATCHAYA M	17	17	44	8	86	Very Good
7	812121101011	ATCHAYA.S	14	14	37	7	72	Good
8	812121101012	BALASUBRAMANIYAN	17	17	42	8	84	Very Good
9	812121101013	BOOBALAN J	19	19	47	9	94	Excellent
10	812121101014	DHANUSH.D	18	18	46	9	91	Excellent
11	812121101015	DHATCHANAMOORTHY A	15	15	38	7	75	Good
12	812121101016	EDWIN JOSWA	18	18	47	9	92	Excellent
13	812121101017	ELAVARASU S	16	16	40	8	80	Very Good
14	812121101018	GOPINATH D	14	14	37	7	72	Good
15	812121101019	HAJI MOHAMED S	17	17	43	8	85	Very Good
16	812121101020	HARISH B	15	15	38	7	75	Good
17	812121101022	JEYA PRIYA.G	18	18	45	9	90	Excellent
18	812121101023	JAYAVEERA PANDI	16	16	40	8	80	Very Good
19	812121101024	JITHENDRAA P	18	18	46	9	91	Excellent
20	812121101025	JOTHIBOSU	14	14	37	7	72	Good
21	812121101026	KALAIYARASAN	16	16	40	8	80	Very Good
22	812121101027	KALIDASS	15	15	38	7	75	Good
23	812121101029	KAVIYARASAN.A	16	16	42	8	82	Very Good
24	812121101030	LEO STANISLAUS	17	17	43	8	85	Very Good
25	812121101031	MANIKANDAN M	16	16	42	8	82	Very Good
26	812121101032	MADHANKUMAR	14	14	36	7	71	Good
27	812121101033	MOHAMMED MEERAN.M	16	16	40	8	80	Very Good
28	812121101034	MUTHU KRISHNAN T	16	16	42	8	82	Very Good
29	812121101035	NAVEEN KUMAR.L	18	18	45	9	90	Excellent



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30	812121101036	NIRANJANEE.R	16	16	42	8	82	Very Good
31	812121101037	NIVETHA V	18	18	46	9	91	Excellent
32	812121101038	OMANA NATHAN.G.L	15	15	37	7	74	Good
33	812121101039	PARTHIBAN B	16	16	40	8	80	Very Good
34	812121101040	POOVARASAN K	16	16	41	8	81	Very Good
35	812121101041	PRAVEENA J	18	18	46	9	91	Excellent
36	812121101043	SAKTHIVEL M	17	17	44	8	86	Very Good
37	812121101044	SATHISH V	14	14	37	7	72	Good
38	812121101045	SELVAKUMAR P	17	17	42	8	84	Very Good
39	812121101046	SHANMUGAM	19	19	47	9	94	Excellent
40	812121101047	SIVAMURUGAN M	18	18	46	9	91	Excellent
41	812121101048	SNEKA.K	17	17	44	8	86	Good
42	812121101050	SYED SABITHULLA S	18	18	47	9	92	Good
43	812121101052	VIGNESH S	16	16	40	8	80	Very Good
44	812121101053	VIJAYALAKSHMI	14	14	37	7	72	Very Good
45	812121101054	VIRUMANDI M	17	17	43	8	85	Excellent
46	812121101055	VISHALI.V	15	15	38	7	75	Very Good
47	812121101056	VISHNU D	19	19	48	9	95	Excellent
48	812121101058	YUVANESHWARAN	16	16	40	8	80	Good

Grade	Marks Range	Count
Poor	Below 60	00
Average	60 - 70	00
Good	70 - 80	13
Very Good	80 - 90	21
Excellent	Above 90	14


Trainer


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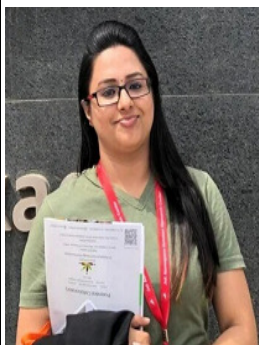
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TRAINER PROFILE

Name	:	Ms.R.S.Krithiga	
Designation	:	Chief Executive Officer	
Highest Qualification	:	M.E Aeronautical Engg	
Field of Interest	:	Drone Simulation	
Company / Institution	:	Jet Aerospace Research Centre	
Contact Number	:	9809812378	
Email ID	:	info.jetaerospace@gmail.com	
Address	:	Near IIT Palakkad I-Hub, Opposite to ITI Ltd, Kanjikode Industrial Area, Palakkad - 678623, Kerala.	
Experience	:	5 years of Training Experience	
Profile	:	She has an experienced and certified drone training instructor with over 5 years of expertise in the field of Drone and Flight Simulator. Specializing in both commercial and recreational drone operations, She is dedicated to providing comprehensive and practical training programs to ensure pilots are proficient, safe, and compliant with all regulatory requirements. • Certified Instructor for DJI Enterprise Products • AUVSI Trusted Operator Program (TOP) Level 3 Certification • Designed and implemented comprehensive training modules for various skill levels. • Conducted over 500 hours of hands-on training sessions.	



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Course Outcome

1. Students were able to define and classify nano satellites and differentiate them from other satellite classes. They explained the historical development and milestones in nano satellite technology.
2. They gained knowledge of satellite mission planning and execution and attained familiarity with launch and deployment options (launch vehicles, deployers)
3. Students understood the satellite systems engineering and integration gained the ability to work in multidisciplinary teams and communicate effectively

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Feedback Form Report

Name of the Program:

VAC Program - Nano Satellites

Date:

12.01.2024

- What is your opinion about the duration of this program?
☐ A. Short ☒ B. Adequate ☐ C. Long
- Overall, how useful was this program for you?
☒ A. Very Much ☐ B. To Some Extent ☐ C. Not Useful
- How would you rate the teaching Qualities?
☐ A. Very Good ☒ B. Good ☐ C. Average ☐ D. Poor
- How would you rate the materials presented?
☐ A. Very Good ☐ B. Good ☒ C. Average ☐ D. Poor
- How much of knowledge you learned today?
☒ A. A Lot ☐ B. Satisfactory ☐ C. None of it
- Did it fulfill your expectation?
☒ A. Yes ☐ B. Some Extent ☐ C. No
- Planning of this programme?
☐ A. Very Good ☒ B. Good ☐ C. Average ☐ D. Poor
- Any other comment (if any):



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VAC Program - Nano Satellites

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☒ A. Very Good ☐ B. Good ☐ C. Average ☐ D. Poor
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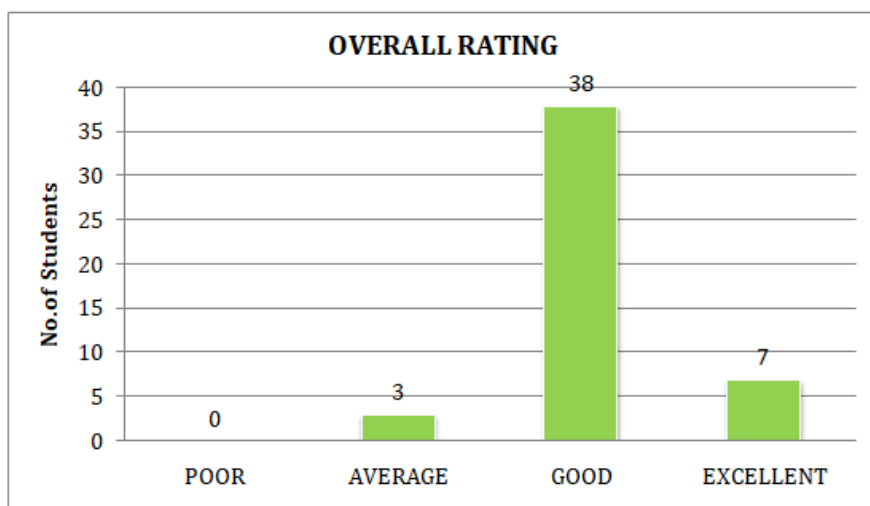
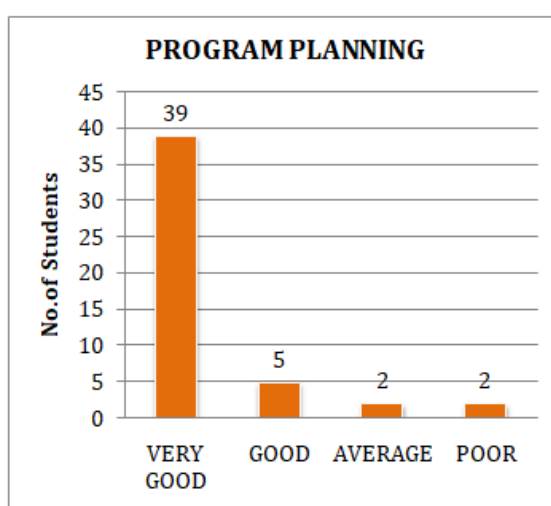
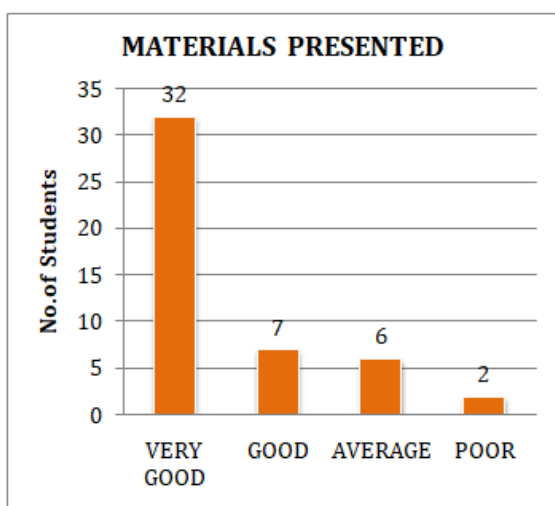
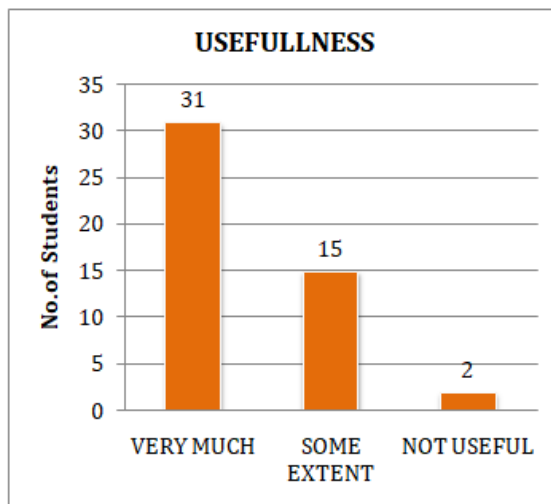
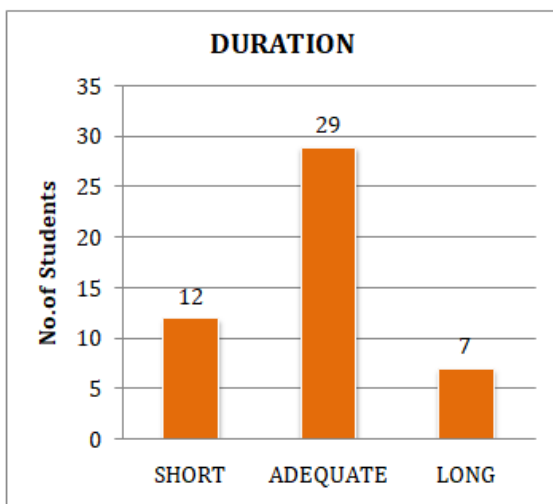
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Feedback Summary





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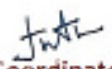


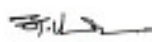
Certificate of Completion

This is to certify that, DHANUSH.D of

III Year Aeronautical Engineering branch has successfully completed the course "*Nano Satellites*" from *08.01.2024* to *12.01.2024* in association with Jet Aerospace

Score: 91 Grade: Excellent


Coordinator


HOD


Principal



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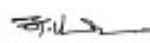
Certificate of Completion

This is to certify that, SELVAKUMAR P of

III Year Aeronautical Engineering branch has successfully completed the course "*Nano Satellites*" from *08.01.2024* to *12.01.2024* in association with Jet Aerospace

Score: 84 Grade: Very Good


Coordinator


HOD


Principal